

Table 3. Unavailable parameters and consequences

Panel A. Evidence status

Missing parameter	Why needed	Status	Sensitivity coordinate allowed	Range status
accessible numerical doxorubicin semiquinone hyperfine tensor	Defines nuclear-spin-driven electron-spin mixing in the exact radical.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter k_doublet_s	Sets doublet-resolved electron-transfer flux.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter k_quartet_s	Sets quartet-resolved electron-transfer flux.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
physical k_quartet_s/k_doublet_s range	Determines reaction selectivity between the two manifolds.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter escape rate	Sets competition between reaction and encounter separation.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter duration	Sets a physical observation endpoint or residence time.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
semiquinone T1	Defines longitudinal relaxation within the encounter.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
semiquinone T2/Tm	Defines coherence decay under a physically matched model.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
exchange coupling	Sets the isotropic electron-electron coupling magnitude.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
dipolar coupling and geometry	Sets anisotropic coupling and requires geometry/orientation.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
solution encounter O2 g and ZFS tensors	Defines encounter-specific oxygen Zeeman/ZFS evolution.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter-specific O2 relaxation	Defines spin relaxation of O2 within the solution encounter.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range

Unavailable inputs remain sensitivity coordinates and limit the conclusions that can be drawn. Presentation values are rounded. All 16 source rows and all 14 source columns remain in table03_unavailable_parameters_and_consequences.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 3. Unavailable parameters and consequences

Panel A. Evidence status (continued)

Missing parameter	Why needed	Status	Sensitivity coordinate allowed	Range status
compartment-specific H2O2 loss rate	Defines compartment-specific removal of downstream H2O2.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter association rate	Connects bulk semiquinone and oxygen concentrations to encounter formation.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
encounter geometry separated from dipolar coupling	Defines separation, orientation, association, and residence physics.	unavailable	yes, with explicit opt-in	illustrative only; no defensible physical range
chemically prepared initial D/Q population	Defines the encounter birth density matrix.	unavailable	yes, as benchmark mixtures only	illustrative only; no measured preparation distribution

Unavailable inputs remain sensitivity coordinates and limit the conclusions that can be drawn. Presentation values are rounded. All 16 source rows and all 14 source columns remain in table03_unavailable_parameters_and_consequences.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 3. Unavailable parameters and consequences

Panel B. Allowed treatment and limitation

Missing parameter	Allowed treatment	Conclusion prohibited	Status source
accessible numerical doxorubicin semiquinone hyperfine tensor	bounded illustrative sensitivity coordinate only	Prohibits an exact-system hyperfine Hamiltonian or mixing rate.	configuration authority and targeted evidence review
encounter k_doublet_s	bounded illustrative sensitivity coordinate only	Prohibits a quantitative D-channel reaction yield.	configuration authority and targeted evidence review
encounter k_quartet_s	bounded illustrative sensitivity coordinate only	Prohibits a quantitative Q-channel reaction yield.	configuration authority and targeted evidence review
physical k_quartet_s/k_doublet_s range	bounded illustrative sensitivity coordinate only	Prohibits assigning a preferred manifold or physical selectivity range.	configuration authority and targeted evidence review
encounter escape rate	bounded illustrative sensitivity coordinate only	Prohibits a quantitative reaction-versus-escape branching ratio.	configuration authority and targeted evidence review
encounter duration	bounded illustrative sensitivity coordinate only	Prohibits a physical encounter time axis and endpoint.	configuration authority and targeted evidence review
semiquinone T1	bounded illustrative sensitivity coordinate only	Prohibits an encounter-specific relaxation rate.	configuration authority and targeted evidence review
semiquinone T2/Tm	bounded illustrative sensitivity coordinate only	Prohibits quantitative dephasing or coherence-lifetime claims.	configuration authority and targeted evidence review
exchange coupling	bounded illustrative sensitivity coordinate only	Prohibits a quantitative exchange Hamiltonian.	configuration authority and targeted evidence review
dipolar coupling and geometry	bounded illustrative sensitivity coordinate only	Prohibits a quantitative dipolar Hamiltonian and orientation average.	configuration authority and targeted evidence review
solution encounter O2 g and ZFS tensors	bounded illustrative sensitivity coordinate only	Prohibits transfer of gas, matrix, or protein O2 tensors into solution.	configuration authority and targeted evidence review
encounter-specific O2 relaxation	bounded illustrative sensitivity coordinate only	Prohibits assigning a solution-encounter O2 relaxation rate.	configuration authority and targeted evidence review

Unavailable inputs remain sensitivity coordinates and limit the conclusions that can be drawn. Presentation values are rounded. All 16 source rows and all 14 source columns remain in table03_unavailable_parameters_and_consequences.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 3. Unavailable parameters and consequences

Panel B. Allowed treatment and limitation (continued)

Missing parameter	Allowed treatment	Conclusion prohibited	Status source
compartment-specific H2O2 loss rate	bounded illustrative sensitivity coordinate only	Prohibits compartment-level H2O2 prediction.	configuration authority and targeted evidence review
encounter association rate	bounded illustrative sensitivity coordinate only	Prohibits a quantitative encounter-formation flux.	references/doxorubicin_parameter_review.md
encounter geometry separated from dipolar coupling	illustrative geometry scenarios only; none used in the baseline	Prohibits a physical association or encounter model.	references/doxorubicin_parameter_review.md
chemically prepared initial D/Q population	unpolarized and manifold-mixture benchmarks only	Prohibits choosing a chemically prepared initial D/Q state.	references/doxorubicin_parameter_review.md

Unavailable inputs remain sensitivity coordinates and limit the conclusions that can be drawn. Presentation values are rounded. All 16 source rows and all 14 source columns remain in table03_unavailable_parameters_and_consequences.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.